Research Article

Pharmacogenomics

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Pharmacogenetic landscape of *DPYD* variants in south Asian populations by integration of genome-scale data

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Aim: Adverse drug reactions to 5-Fluorouracil(5-FU) is frequent and largely attributable to genetic variations in the *DPYD* gene, a rate limiting enzyme that clears 5-FU. The study aims at understanding the pharmacogenetic landscape of DPYD variants in south Asian populations. **Materials & Methods:** Systematic analysis of population scale genome wide datasets of over 3000 south Asians was performed. Independent evaluation was performed in a small cohort of patients. **Results:** Our analysis revealed significant differences in the the allelic distribution of variants in different ethnicities. **Conclusions:** This is the first and largest genetic map the DPYD variants associated with adverse drug reaction to 5-FU in south Asian population. Our study highlights ethnic differences in allelic frequencies.

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The 5-fluorouracil (5-FU) is an antimetabolite drug [1] widely prescribed for chemotherapy of solid tumors [2]. The antineoplastic drug [3] is used alone or in combination with other drugs and is a mainstay for gastrointestinal cancer [4–10]. It is estimated that nearly 40% of cancer patients administered with 5-FU develop toxicity-associated symptoms [11]. These symptoms could range from hand–foot syndrome, mucositis, granulocytopenia, neuropathy, to severe adverse events or even death in some individuals [12].

Within the cell, 5-FU is converted into multiple active metabolites through a series of enzymatic reactions. One of the active derivatives of 5-FU is fluorodeoxyuridine monophosphate, a competitive inhibitor of thymidylate synthase, which blocks the conversion of deoxyuridine monophosphate to deoxythymidine monophosphate [1,12–13], a crucial precursor of thymine. This results in an imbalance in nucleotide availability, crucial for rapidly dividing cancer cells [14], which leads to imbalanced cell growth and eventually cell death [12]. Additionally being a pyrimidine analog, 5-FU can incorporate into DNA and RNA, eliciting DNA-damage repair and triggering apoptosis machinery [4].

More than 80% of the administered 5-FU is converted to α-fluoro-β-alanine, which is the end point of the catabolic schemes of 5-FU [15]. Dihydropyrimidine dehydrogenase (DPD) is the first enzyme in this catabolic process that leads to clearance of 5-FU [2]. DPD is also the rate-limiting enzyme in the catabolism of 5-FU, which takes place in liver wherein DPD is abundantly expressed [1,16]. Hence, the bioavailability of 5-FU is mainly regulated by DPD, thus determining its half-life, efficacy and toxicity in treatment [2,13,17]. The human *DPYD* gene (NM_000110)

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is highly conserved in mammalian evolution and is located on chromosome 1p22 (chr1:97,077,744–97,921,059), which is 843,316 bp long containing 23 exons [18,19]. The highly polymorphic *DPYD* gene encodes DPD, an enzyme made of 1025 amino acids [20].

Partial or complete deficiency of DPD could result in adverse drug reactions following administration of 5-FU [17]. It is estimated that as much as 5% of the world's population is genetically predisposed to DPD deficiency [21–23], making them susceptible to 5-FU toxicity upon exposure [24]. Over 7000 genetic variations have been reported in *DPYD* gene spanning both the coding and noncoding regions [25]. Though most are likely to be harmless, a few are expected to be responsible for the altered DPD function between individuals [25]. While a few of the variants like the c.1905+1G>A, c.2846 A>T and c.1679 T>G in homozygous state have been known to cause loss of DPD function [19,25] resulting in familial pyrimidinemia, the partial deficiency in heterozygous condition also causes severe 5-FU-associated toxicity [26] upon exposure.

The chemotherapeutic drug 5-FU is also a well-known treatment option for dermatological conditions like actinic keratosis, superficial basal cell carcinoma, recalcitrant hypertrophics and keloid lesions [27,28]. As many as 13 fluorouracil drugs are in prescription as per the US FDA guidelines, of which more than 60% are in injectable form with 50 mg/ml strength and the rest are available as topical cream with 0.5–5% strength. Fluorouracil cream is prescribed for multiple actinic or solar keratoses of the face and anterior scalp while the injections are for treatment of various neoplasms [29,30]. FDA drug label for both the cream and injection cautions about the probable drug adversity in cases with DPD deficiency.

Two previous studies, Saif *et al.* [31] and Iyer *et al.* [32] have profiled c.1905+1G>A variant frequency among small Indian cohorts (225 healthy volunteers). This was followed by a more comprehensive profiling by Iyer *et al.*, 2015 [33] where 50 individuals from India were analyzed for variants in the *DPYD* gene through targeted sequencing. Both these studies provided an allele frequency estimate for as many as 22 genetic variants, though limited by sample numbers.

In the present study, we integrated genome-scale data from 3140 samples belonging to multiple cohorts (n = 16) to ascertain the allele frequencies of variants of pharmacogenetic relevance to 5-FU in *DPYD* gene. We also did clinical validation of these high confidence pharmacogenetic variants in a cohort of 110 cancer patients undergoing chemotherapy. To the best of our knowledge, this is the most comprehensive analysis of a large cohort of individuals providing the baseline demography of pharmacogenetic variants in *DPYD* for 5-FU in any population/in south Asian population.

Materials & methods

Study population & datasets

We integrated a number of datasets from disparate studies and platforms ranging from genotyping microarray to whole exome and whole genome sequencing representing south Asians. This integrated data of south Asian Genomes and Exomes (SAGE) was used for all our analysis. We obtained the genotype information in variant call format for the study population. The summary of the SAGE datasets and respective studies are summarized below. The details of the data are summarized as Table 1.

Wellness genome project

This in-house dataset consists of 93 samples with 63 whole exomes and 30 genomes of healthy Indians who are above the age of 90. The exome capture was performed using Truseq exome enrichment.

Indian Diabetes Consortium

This dataset included 1915 individuals who were genotyped in Indian Diabetes Consortium (INDICO), a cross-sectional cohort of individuals studied for genetic markers associated with Type 2 diabetes mellitus. The cohort of samples were derived from five states (Punjab, Haryana, Delhi, Uttar Pradesh and Bihar) in north India. They belonged to Indo-European ancestry and their age ranged from 29 to 80 years. The dataset comprised of genotype data generated on Illumina platform (Illumina Inc., CA, USA) as published previously [35,40]. The raw genotypes were filtered as per quality criteria described in the publication. The final filtered SNPs had an average call rate of 98.2% and concordance rate of over 99%.

Table 1. List of projects, the population and the cohort size of the study in south Asian Genomes and Exome.

Project	Code	Cohort size	WGS	WES	Description	Number of genetic variants	Ref.
1000G SAS	PJL	158	108		Punjabi from Lahore, Pakistan	120,801,880	[34]
	STU	128	111		Sri Lankan Tamil from the UK		
	GIH	113	109		Gujarati Indian from Houston, Texas		
	BEB	144	103		Bengali from Bangladesh		
	ITU	118	113		Indian Telugu from the UK		
INDICO	BIH	268	Genotype frequencies		Indian Diabetes Consortium	616,795	[35]
	PUN	225					
	UPR	1102					
	HAR	109					
	DEL	213					
Southeast Asian Malays	SSMP	100	100	0	Cosmopolitan Malays from Singapore	15,632,867	[36]
The south Asian Genome	SAG	321	174	147	South Asian Genome	12,977,578	[37]
StatGen	SSIP	38	38	0	Indians living in Singapore	11,555,478	[38]
Wellness Genome project	WGP	93	63	30	Genomes of the Centenarians	48,776,586	In-house data
Andaman Genomes	PGA	10	10	0	Population Genetics of Andamanese	15,254,294	[39]
Total	SAGE	3140	931	177	South Asians Genomes and Exomes		

BEB: Bengali in Bangladesh; BIH: Bihar; DEL: Delhi; GIH: Gujarati in Houston, UK; HAR: Haryana; ITU: Indian Telugu in UK; PGA: Population Genetics of Andamanese; PJL: Punjabi in Lahore; PUN: Punjab; SAG: South Asian genomes; SAGE: South Asian genomes and exomes; SAS: South Asians; SSIP: Singapore Sequencing Indian Project; STU: Srilankan Tamil in UK; UPR: Uttar Pradesh; WGP: Wellness Genome Project.

South Asian Indian Genomes

The dataset of south Asian Indian genomes and exomes encompassing a total of 321 individuals were derived from previously published reports. The dataset comprises 174 whole genomes (4.3× coverage – 168 samples and 28.4× coverage – 8 samples) and 147 whole exome data (20.6× coverage) [37]. Sequencing was performed in Illumina HiSeq 2000. To ensure the quality of the data, initial set of SNPs and indels called were recalibrated using dbSNP as truth set for machine learning and filtered to provide SNP and indel call sets with predicted positive values of 99.9 and 95%, respectively.

South Asian samples from 1000 Genomes Project Data

The allele frequencies of south Asian samples from the 1000 Genomes Project data were derived from the 1000 Genome consortium database. The dataset encompasses 84,801,880 genomic variations from 3511 individuals. The individuals belonged to 27 populations containing 1760 females and 1740 males. Of this 351 males and 310 females represent south Asian origin contributing to a total of 661 samples from within five subpopulations. The populations include Gujarati Indian from Houston, Texas; Punjabi from Lahore, Pakistan; Bengali from Bangladesh; Sri Lankan Tamil from the UK and Indian Telugu from the UK which encompass a total of 661 genomes [34].

Singapore Sequencing Indian Project & Singapore Sequencing Malay Project

Allele frequencies for variants were also analyzed from the whole genome dataset of 38 individuals of south Indian ancestry from Singapore with 12 males and 26 females [38]. Illumina HiSeq 2000 was used for sequencing to achieve a target coverage of 30×. Only variants that were identified by both CASAVA and GATK were reported.

Similarly the allele frequencies were also derived from the SSMP cohort of a total of 100 cosmopolitan Malays (50 males and 50 females) living in Singapore [36]. Malay samples were sequenced with a minimum of 30× depth coverage in Illumina HiSeq 2000. Only the consensus set of variants identified by CASAVA and SAMtools were reported.

Population genetics of Andamanese (PGA)

Population genetics of Andamanese (PGA) consists of genomes from ten individuals from the Jarawa and Onge populations in the Andaman Islands from previously published reports [39]. HiSeq 2000 platform was used for sequencing with 15× coverage. To ensure the quality of data, the differences in coverage among individuals were checked through depth of coverage from GATK. Transition versus transversion ratio was also checked to ensure the quality of the data.

Allele frequencies & genotype frequencies

We calculated the allele and genotype frequencies for the variants of our interest in the study population. We also compared the allele frequencies in the integrated dataset vis-a-vis global allele frequency datasets. Allele frequencies for global population for comparison was derived from the 1000 Genomes [34], ExAC [41] and HapMap [42]. The variant datasets were obtained in variant call format generated through Phase III pipeline (based on human reference assembly GrCh37). Minor allele frequencies (a) for each population were calculated using bespoke python scripts as:

$$f(a) = \frac{Aa + 2\times(aa)}{2\times(AA) + 2\times(Aa) + 2\times(aa)}$$

While the genotype frequency (aa) was calculated as:

$$f(aa) = \frac{aa}{AA + Aa + aa}$$

Computational annotation of novel variants

Functional annotation of the variants with dbSNP, refSeq was performed using ANNOVAR. Variants not reported previously in dbSNP were classified as novel variants. All the nonsynonymous variants were evaluated using a battery of three computational algorithms SIFT [43], PolyPhen-2 [44] and PROVEAN [45] for predicting potentially deleterious variations.

Selection of genetic variants

The variants in *DPYD* gene were selected based on the clinical annotation level of evidence in PharmGKB [46], a comprehensive database of pharmacogenetic variants. There were 26 variants that were distributed across three evidence levels (1A, 3 and 4). We chose 15 variants that were associated with 5-FU, among which three variants viz., rs67376798 (c.2846 A>T), rs55886062 (c.1679 T>G) and rs3918290 (**2A*) belonged to the 1A evidence level having supported by a number of studies. The variant rs2297595(c.496A>G) that was previously annotated as level 2A and currently in level 3 along with other 11 variants rs75017182, rs1801265, rs1801160, rs1801159, rs1801158, rs1760217, rs17376848, rs12132152, rs12022243, rs115232898 and rs76387818 were chosen for further analysis in the study. The variants are summarized in Supplementary Table 1 along with their level of evidence.

Statistical analysis

All single nucleotide variants used in the present study were tested for Hardy–Weinberg equilibrium (HWE) in the study population. Observed genotype frequencies were compared with those expected under HWE using the χ^2 test. south Asian allele frequencies were compared with that of the 1000 Genomes Project (Phase III) global population average (ALL) using Fisher's Exact test. The distribution of allele frequencies among various populations were visualized by a scatter plot in Python's matplotlib package [47].

Clinical validation

Human subjects

The purpose of this study was to assess the analytical validity of our computational prediction from healthy south Asian controls in patients undergoing chemotherapy. This study was approved by the Institutional Human Ethics Committee. A total of 110 patients on chemotherapy with 5-FU presenting to a tertiary care center in India were enrolled for the study. All patients provided written informed consent for the study. Clinical information was collected from medical records, including age, sex, ethnicity, weight and medications. Five milliliter of whole blood was collected in vacutainers from patients and was used for DNA isolation followed by PCR and sequencing.

PCR amplification & Sanger sequencing

Polymerase chain reaction based amplicons were generated for Sanger sequencing to capture 15 *DPYD* variants. The list of primers used for amplification is summarized in Supplementary Table 2. PCR was performed in a total reaction volume of 25 μl with GOtaq colorless master mix (Promega, WI, USA) with respective primers (0.4 pm) and template DNA (100 ng). PCR conditions were similar for all the primer pairs, initial denaturation of 95°C for 5 min and cycling conditions of 95°C for 30 s, annealing Tm 60°C for 30 s and final extension at 72°C for 35 cycles. PCR amplicons were sequenced by Sanger method using forward and reverse primers and the sequences were aligned and analyzed by Seqman (DNASTAR) [48] for variant identification.

Statistical analysis

The side effects of the patients were dichotomized as symptomatic and asymptomatic and the odds ratio for the side effect with 95% CI was calculated. Similarly odds ratio was also calculated for adverse effect (grade 3 and above) versus no effect. The patients were grouped into three groups for the odds ratio calculation, individuals with heterozygous genotype (aA), homozygous genotype (aa) and either aa or aA. Bespoke python scripts were used for the computation.

Results

Allele & genotype frequencies in the south Asian datasets

Genotype information for analysis were available for 661(rs67376798, rs55886062, rs115232898), 1084 (rs76387818), 1215(rs75017182), 1225 (rs12022243), 1122(rs1801158), 1215(rs1801160), 2707 (rs3918290), 2999 (rs12132152), 3092 (rs17376848), 3140 (rs1801159, rs1760217, rs1801265 and rs2297595) individuals for each of the variants considered in the present analysis. Of the 15 SNPs analyzed, 10 of them cause a single amino acid change with rs1801160 causing a synonymous change in the protein and three intronic variants (rs75017182, rs1760217 and rs12022243) while rs3918290 is a splice donor variant. Variants rs1801265 and rs12022243 showed higher frequency among the variants studied in south Asian populations. Interestingly rs1801265 shows a higher frequency in the homozygous form. Variants rs115232898, rs55886062, rs67376798 are almost not present in the study population. Allele frequency and genotype frequency is summarized in Tables 2 & Table 3, respectively.

The genotype frequencies for all variants were compared and departure from HWE was tested. Expected genotype frequencies of variants were calculated by Hardy–Weinberg law in the study population. The homozygous variant for rs67376798, rs55886062, rs3918290, rs75017182, rs1801158, rs17376848, rs12132152, rs115232898 and rs76387818 are absent in almost all of the populations studied. Among selected variants, the rs1801265 homozygous variant genotype displayed the highest frequency, ranging from 0.42 to 0.61 and the heterozygous variant genotype displayed a frequency range from 0.35 to 0.51 in the study population. Variant rs12022243 in the heterozygous genotype display highest frequency range from 0.41 to 0.46 in the study population.

Comparison of allele frequencies with global populations

We analyzed the shortlisted variants and compared with the allele frequencies across the global population. We have used data from 1000 Genomes project, ExAC, ESP500 and HapMap projects for the global population. The population frequencies of alleles in different datasets were computed for each population. We used Fisher's Exact test to compare the differences in allele frequencies of the 15 pharmacogenetic variants of *DPYD* gene between the south Asians and the ALL of 1000 genomes. Figure 1 shows the allele frequency distribution among Indian and global population.

Table 2. Allele frequencies of the dihydropyrimidine dehydrogenase variants in the datasets considered.

dbSNP_ID	AA change	SAS	INDICO	SSMP	SAG	SSIP	WGP	PGA
rs67376798	Asp→Val	T: 0.999 A: 0.001	NA	NA	NA	NA	NA	NA
rs55886062	Ile→Asn	A: 1 C: 0	NA	NA	NA	NA	NA	NA
rs3918290	Splice	C: 0.992 T: 0.008	C: 0.998 T: 0.002	NA	NA	C: 0.986 T: 0.014	NA	NA
rs2297595	Met→Val	C: 0.938 T: 0.062	C: 0.913 T: 0.087	C: 0.908 T: 0.092	C: 0.908 T: 0.092	C: 0.917 T: 0.083	C: 0.924 T: 0.077	C: 0.9 T: 0.1
rs75017182	Intronic	G: 0.98 C: 0.020	NA	G: 0.979 C: 0.021	G: 0.979 C: 0.021	G: 0.986 C: 0.014	G: 0.889** C: 0.111	NA
rs1801265	Cys→Arg	G: 0.266 A: 0.734	G:0.243** A: 0.757	G: 0.233 A: 0.767	G: 0.223 A: 0.777	G: 0.208 A: 0.792	G:0.219** A: 0.782	G: 0** A: 1
rs1801160	Val→Ile	C: 0.913 T: 0.087	NA	C: 0.908 T: 0.092	C: 0.898 T: 0.102	C: 0.930 T: 0.070	C: 0.875* T: 0.125	NA
rs1801159	Ile→Val	T: 0.917* C: 0.083	T: 0.917* C: 0.083	T: 0.911* C: 0.089	T: 0.912* C: 0.088	T: 0.917* C: 0.083	T: 0.923* C: 0.077	T: 0.95** C: 0.05
rs1801158	Ser→Asn	C: 0.995 T: 0.005	NA	C: 0.995 T: 0.005	C: 0.987 T: 0.013	C: 0.972 T: 0.028	NA	NA
rs1760217	Intronic	A: 0.85 G: 0.150	A: 0.839 G: 0.161	A: 0.839 G: 0.161	A: 0.844 G: 0.156	A: 0.833 G: 0.167	A: 0.833 G: 0.167	A: 0.65* G: 0.35
rs17376848	Phe→Phe	A: 0.965 G: 0.035	A: 0.970 G: 0.030	A: 0.955 G: 0.045	A: 0.960 G: 0.040	NA	A: 0.960 G: 0.040	NA
rs12132152	Intergenic	G: 0.995 A: 0.005	G: 0.995 A: 0.005	G: 0.994 A: 0.006	G: 0.994 A: 0.006	NA	NA	NA
rs12022243	Intronic	C: 0.594** T: 0.406	NA	C: 0.664* T: 0.336	C: 0.668* T: 0.332	C: 0.542** T: 0.458	C: 0.5** T: 0.5	C: 0.65** T: 0.35
rs115232898	Tyr→Cys	T: 1 C: 0	NA	NA	NA	NA	NA	NA
rs76387818	Downstream gene variant	G: 0.995 A: 0.005	NA	G: 0.994 A: 0.006	G: 0.994 A: 0.006	NA	NA	NA

*Indicates the p-value < 0.05 in Fisher's Exact test.
**Indicates p-value < 0.01 in Fisher's Exact test.
AA:Amino acid; INDICO: Indian Diabetes Consortium; PGA: Population genetics of Andamanese; NA: Not available; SAG: South Asian genomes; SAS: South Asians; SSIP: Singapore Sequencing Indian Project; SSMP: Singapore Sequencing Malay Project; WPG: Wellness Genome Project.

Variant classification in DPYD gene

We further analyzed the variome for novel variants in the *DPYD* gene. We found that among the 27,397 variants of *DPYD* gene, 17,067 variants are novel and previously not reported in dbSNP. A total of 27,273 are present in the nonprotein coding region of the gene. There are 108 variants present in the protein coding region of which 83 variants cause single amino acid change (shown in the Figure 2), 23 synonymous (all of them are neutral variants with a PROVEAN score = 0) and two nonsense variations. A total of 65 variants are reported as deleterious and 36 as neutral variants by the PROVEAN prediction. While SIFT predicted 38 single amino acid change as tolerated and 45 as damaging, Polyphen-2 predicts 39 as probably damaging and 42 variants as benign.

Clinical validation on a limited cohort of patients

Clinical validation was done on a limited cohort of 110 cancer patients. The individuals of this cohort belonged to an age group of 15–82 with 35 females and 75 males having either of colon, stomach, esophageal, gastrointestinal or rectal cancer. Patient characteristics are summarized in Table 4. These patients were undergoing chemotherapy ranging from cycle 1 to cycle 7 with either of FOLFOX, CAPOX, capecitabine, paclitaxel + carboplatin and 5-FU/Irinotecan. A total of 70 percent of individuals had adverse effects with the therapy. In our patient cohort, variants rs12022243 and rs115232898 were absent. One patient was found to be heterozygous for the variant rs76387818 and one for rs1801158. Two patients were found to be heterozygous for rs3918290 and these patients had grade III HFS and diarrhea. A total of 19 out of 110 patients were positive for rs2297595 and of which 17 were heterozygous (CT) and 2 were homozygous (CC). Table 5 shows the genotype frequency distribution of the patient cohort. Table 6 shows the side effect observed in patients undergoing chemotherapy with 5-FU and the odds of having

Table 3. Genotype frequencies of the dihydropyrimidine dehydrogenase variants in the datasets considered.							
dbSNP_ID	**SAS**	**INDICO**	**SSMP**	**SAG**	**SSIP**	**WGP**	**PGA**
Rs67376798	TT: 1 AT: 0 AA: 0	NA	NA	NA	NA	NA	NA
Rs55886062	AA: 1 AC: 0 CC: 0	NA	NA	NA	NA	NA	NA
Rs3918290	CC: 0.99 CT: 0.01 TT: 0	CC: 1.00 CT: 0 TT: 0	NA	NA	CC: 0.97 CT: 0.03 TT: 0	NA	NA
rs2297595	TT: 0.88 CT: 0.11 CC: 0.01	TT: 0.83 CT: 0.16 CC: 0	TT: 0.83 CT: 0.15 CC: 0.02	TT: 0.83 CT: 0.15 CC: 0.02	TT: 0.83 CT: 0.17 CC: 0	TT: 0.86 CT: 0.12 CC: 0.02	TT: 0.8 CT: 0.2 CC: 0
rs75017182	GG: 0.99 CG: 0.01 CC: 0	NA	GG: 0.96 CG: 0.04 CC: 0	GG: 0.96 CG: 0.04 CC: 0	GG: 0.97 CG: 0.03 CC: 0	GG: 0.78 CG: 0.22 CC: 0	NA
rs1801265	GG: 0.08 AG: 0.40 AA: 0.52	GG: 0.06 AG: 0.36 AA: 0.58	GG: 0.06 AG: 0.35 AA: 0.60	GG: 0.05 AG: 0.35 AA: 0.60	GG: 0.03 AG: 0.36 AA: 0.61	GG: 0.03 AG: 0.37 AA: 0.60	GG: 0 AG: 0 AA: 1
rs1801160	CC: 0.85 CT: 0.15 TT: 0.01	NA	CC: 0.83 CT: 0.16 TT: 0.01	CC: 0.81 CT: 0.18 TT: 0.01	CC: 0.86 CT: 0.14 TT: 0	CC: 0.76 CT: 0.22 TT: 0.01	NA
rs1801159	TT: 0.85 TC: 0.15 CC: 0.01	TT: 0.84 TC: 0.16 CC: 0.01	TT: 0.84 TC: 0.14 CC: 0.02	TT: 0.83 TC: 0.16 CC: 0.01	TT: 0.83 TC: 0.17 CC: 0	TT: 0.85 TC: 0.15 CC: 0	TT: 0.9 TC: 0.1 CC: 0
rs1801158	TT: 1.00 TC: 0 CC: 0	NA	TT: 0.99 TC: 0.01 CC: 0	TT: 0.98 TC: 0.02 CC: 0	TT: 0.94 TC: 0.06 CC: 0	NA	NA
rs1760217	AA: 0.73 AG: 0.25 GG: 0.02	AA: 0.71 AG: 0.27 GG: 0.03	AA: 0.70 AG: 0.29 GG: 0.02	AA: 0.70 AG: 0.28 GG: 0.02	AA: 0.72 AG: 0.22 GG: 0.06	AA: 0.67 AG: 0.33 GG: 0	AA: 0.4 AG: 0.5 GG: 0.1
rs17376848	AA: 0.94 AG: 0.06 GG: 0	AA: 0.94 AG: 0.06 GG: 0	AA: 0.91 AG: 0.09 GG: 0	AA: 0.92 AG: 0.08 GG: 0	NA	AA: 0.92 AG: 0.08 GG: 0	NA
rs12132152	GG: 0.99 AG: 0.01 AA: 0	GG: 0.99 AG: 0.01 AA: 0	GG: 0.99 AG: 0.01 AA: 0	GG: 0.99 AG: 0.01 AA: 0	NA	NA	NA
rs12022243	CC: 0.42 CT: 0.41 TT: 0.17	NA	CC: 0.43 CT: 0.46 TT: 0.11	CC: 0.44 CT: 0.46 TT: 0.10	CC: 0.33 CT: 0.42 TT: 0.25	CC: 0.17 CT: 0.65 TT: 0.17	CC: 0.4 CT: 0.5 TT: 0.1
rs115232898	TT: 0.99 CT: 0.01 CC: 0	NA	NA	NA	NA	NA	NA
rs76387818	GG: 0.99 AG: 0.01 AA: 0	NA	GG: 0.99 AG: 0.01 AA: 0	GG: 0.99 AG: 0.01 AA: 0	NA	NA	NA

INDICO: Indian Diabetes Consortium; PGA: Population Genetics of Andamanese; NA: Not Available; SAG: South Asian Genomes; SAS: South Asians; SSIP: Singapore Sequencing Indian Project; SSMP: Singapore Sequencing Malay Project; WPG: Wellness Genome Project.

the side effect in presence of the genotype. Table 7 shows the adverse effect and its odds to the 5-FU uptake. We found that the odds of having neuropathy and hand–foot syndrome, the classical symptoms of 5-FU toxicity were observed in patients having rs2297595 variant (Table 5). We also observed that hand–foot syndrome was found in grade III and above in patients with variant rs2297595. The other variants that were positive in the cohort did not have significant correlation with any of the classical symptoms of 5-FU toxicity (Supplementary Table 4).

Discussion

Various types of solid and malignant tumors are treated with fluoropyrimidines like 5-FU [49]. Being an uracil analog 5-FU exploits its transport mechanism to rapidly enter the cell [1]. Once inside the cell this antineoplastic compound inhibits thymidylate synthase and thereby interfering effective DNA synthesis [50,51]. Although 5-FU is used for chemotherapy of various cancers for more than five decades, this generally well-tolerated drug has reports of magnitudes of toxicity by nearly 40% of the patients [52]. Difference in the clearance rate of 5-FU contributes

10.2217/pgs-2017-0101

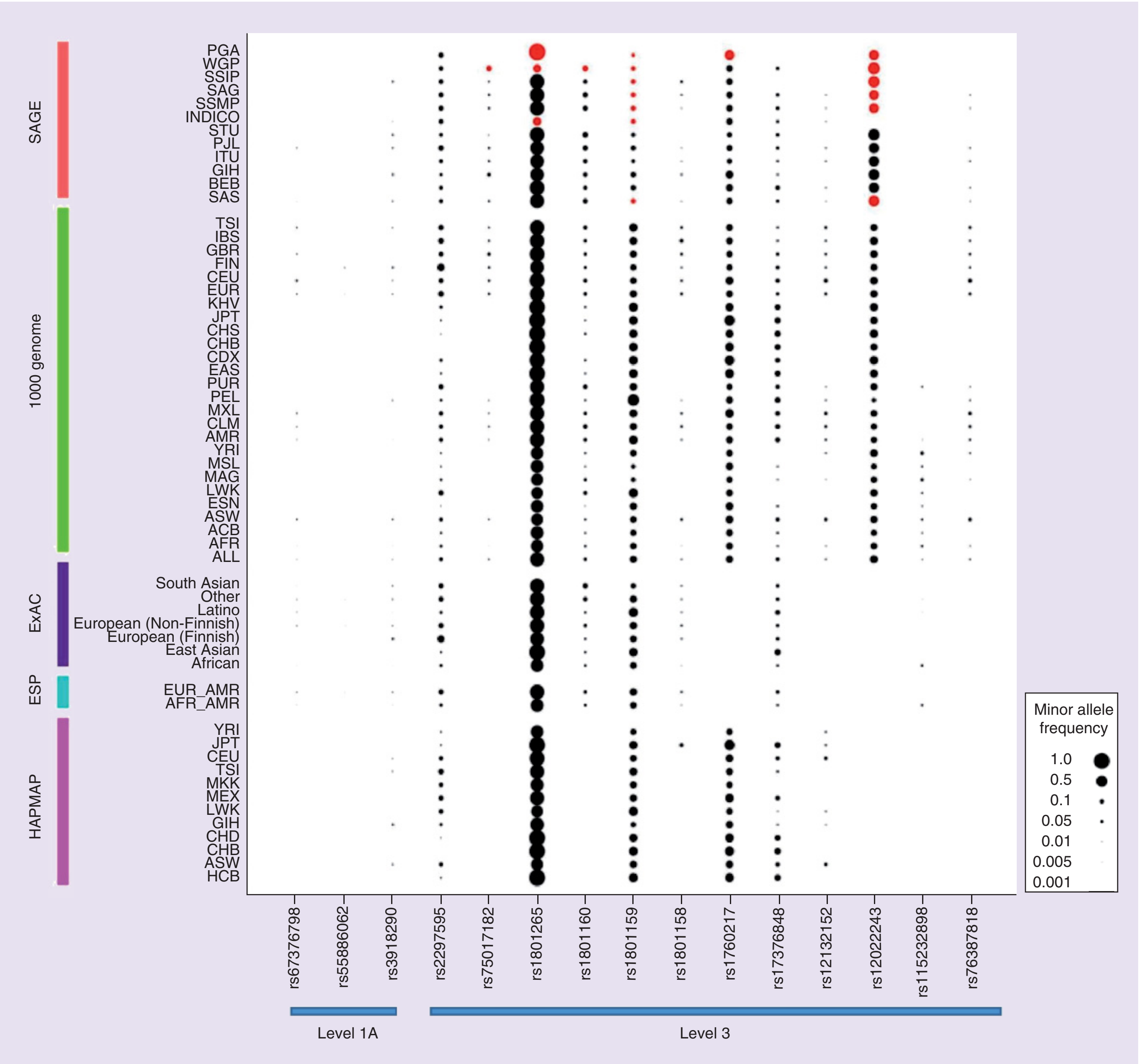

Figure 1. Allele frequency distribution among all the south Asian population. The figure shows a bubble plot of the distribution of allele frequencies of *DPYD* genetic variants. Size of the bubble represents the minor allele frequency. The significant allele frequency with respect to the global population are marked in red color. The blue line in the figure marks the level of evidence of the variant in PharmGKB database.
1000 Genome south Asian genome.
AFR: African; AMR: American; BEB: Bengalis in Bangladesh; EAS: East Asian; EUR: European; GIH: Gujarati in Houston; INDICO: Indian Diabetes Consortium; ITU: Indian Telugu in UK; PGA: Population genetics of Andaman; PJL: Punjabi in Lahore; SAG: South Asian genome; SAGE: South Asian genomes and exomes; SAS: South Asian; SSMP: Southeast Asian Malays; STU: Srilankan Tamilian in UK.

to varying degrees of toxicity in patients undergoing 5-FU treatment. More than 80% of the administered drug is purged from the system by *DPYD* [53]. Mutated *DPYD* is found in various populations and patients with the same have reported toxicities while receiving 5-FU [54,55].

The FDA recommends patients with DPD enzyme deficiency to not use 5-FU drugs like carac, adrucil, efudex and others as severe toxicity is observed in such patients. FDA also reports that despite meticulous selection of

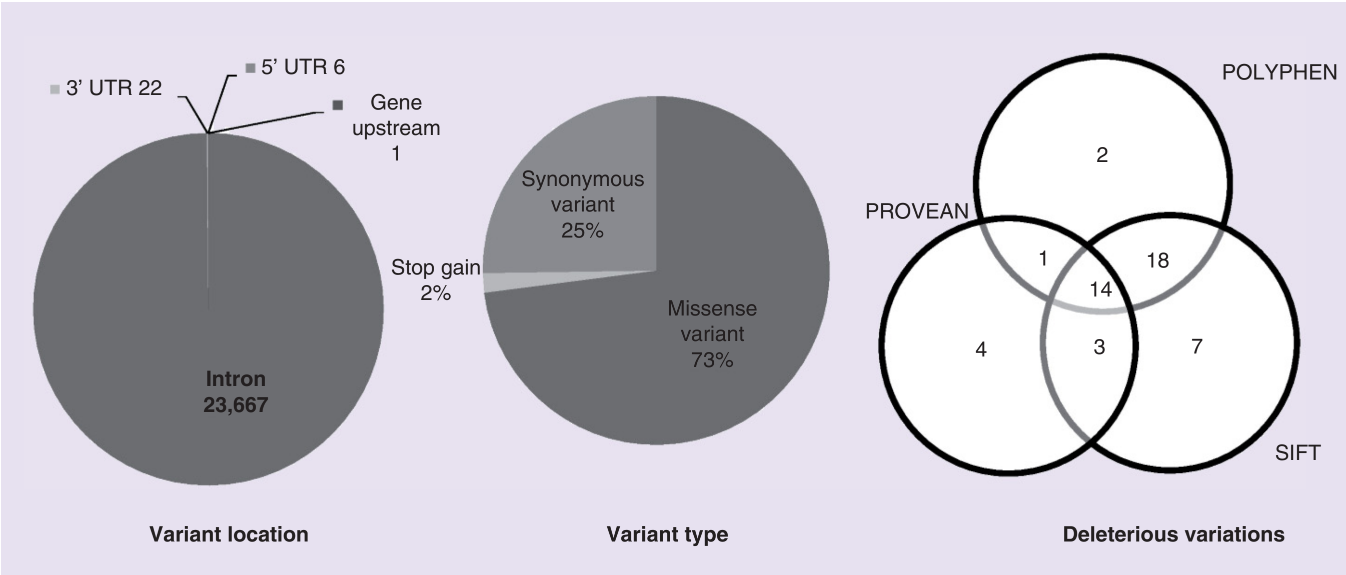

Figure 2. Variant annotation. Variant annotation revealing the position of the variants in the genomic region. The type of variant. Venn diagram depicting the common deleterious variants annotated by SIFT, PolyPhen and Provean database.

Table 4. Patient characteristics of the validation cohort.

Characteristic	Value (number)
Median age	60 years
Sex ratio	
Male	75
Female	35
Site of cancer	
Colon	36
Rectal	41
Stomach	9
Esophagus	7
Gastroesophageal junction	6
Others	11
Regimens used	
Capecitabine	19
FOLFOX	20
CAPOX	38
EOX	4
FOLFIRI	5
IFL	2
Combination (others)	22
Median chemo cycle	3

CAPOX: Capecitabine + oxaliplatin; EOX: Epirubicin + oxaliplatin + capecitabine; FOLFIRI: Folinic acid + fluorouracil + irinotecan; FOLFOX: Folinic acid + fluorouracil + oxaliplatin; IFL: Irinotecan,+ folinic acid +fluorouracil (bolus injection).

patients and careful adjustment of dosage, use of 5-FU might result in toxicity. Prompt discontinuation of 5-FU is recommended in patients exhibiting signs of toxicity like stomatitis, hand–foot syndrome, leukopenia, diarrhea, gastrointestinal ulceration and bleeding or hemorrhage from any site.

Inherited factors and noninherited causes contribute largely in making an individual tolerant or susceptible toward a drug and it considerably varies among ethnic groups. In an ethnically stratified population like south Asia,

Table 5. Genotype frequency distribution of DPYD variants in the clinical study.

SNPs	rs-67376798	rs-55886062	rs-3918290	rs-2297595	rs-75017182	rs-1801265	rs-1801160	rs-1801159	rs-1801158	rs-1760217	rs-17376848	rs-12132152	rs-12022243	rs-115232898	rs-76387818
Homozygous Ref_allele	TT: 1	AA: 1	CC: 0.97	TT: 0.83	GG: 0.96	NW	CC: 0.76	TT: 0.88	TT: 0.99	AA: 0.81	AA: 0.91	GG: 0.82	CC: 1	TT: 1	GG: 0.99
Heterozygous	AT: 0	AC: 0	CT: 0.03	CT: 0.15	CG: 0.04	NW	CT: 0.21	TC: 0.12	TC: 0.01	AG: 0.18	AG: 0.09	AG: 0.17	CT: 0	CT: 0	AG: 0.01
Homozygous Alt_allele	AA: 0	CC: 0	TT: 0	CC: 0.02	CC: 0	NW	TT: 0.03	CC: 0.00	CC: 0	GG: 0.02	GG: 0	AA: 0.01	TT: 0	CC: 0	AA: 0

NW: Not working.

Table 6. Odds ratio with 95% CI of variant rs2297595 for side effects in patients carrying the genotype CT, TT and CT+TT.

Symptomatic_versus_asymptomatic									
	CT			TT			CT+TT		
Side effect	OR	p-val	95% CI	OR	p-val	95% CI	OR	p-val	95% CI
Neuropathy	3.5893	0.0399	1.0605–12.1481	4.1875	0.3204	0.2484–70.6054	3.6641	0.0271	1.1582–11.5916
HFS	4.64	0.0196	1.2790–16.8326	6.8824	0.2424	0.2711–174.7303	5.22	0.0106	1.4693–18.5455
Mucositosis	2.8125	0.1714	0.6390–12.3790	26.6471	0.0497	1.0044–706.9472	3.75	0.0586	0.9532–14.7526
Neutropenia	0.7917	0.7787	0.1552–4.0384	Na	Na	Na	0.7917	0.7787	0.1552–4.0384
GCSF	0.7166	0.6827	0.1450–3.5419	11.5714	0.1390	0.4516–296.5162	1.0749	0.9185	0.2696–4.2859
Toxicity	0.75	0.7947	0.0859–6.5481	25.5882	0.0526	0.9642–679.0746	1.5	0.6333	0.2836–7.9339
Diarrhea	2.6296	0.1533	0.6974–9.9150	1.9067	0.6977	0.0734–49.4983	2.3667	0.1978	0.6379–8.7803

GCSF: Granulocyte-colony stimulating factor requirement; HFS: Hand and Foot Syndrome; Na: Not Available.

Table 7. Odds ratio with 95% CI of variant rs2297595 for adverse effect in patients carrying the genotype CT, TT and CT+TT.

Adverse effect versus_no effect									
	CT			TT			CT+TT		
Side effect	OR	p-value	95% CI	OR	p-val	95% CI	OR	p-value	95% CI
Neuropathy	1.2857	0.872	0.0604–27.3599	22.3333	0.0427	1.1080–450.1541	2.7917	0.0141	2.3629–2133.3722
HFS	7.25	0.0237	1.3024–40.3589	9	0.282	0.1644–492.8028	7.25	0.3977	0.2586–30.1337
Mucositis	7.1905	0.3304	0.1353–382.0527	151	0.0408	1.2349–18463.9586	7.1905	0.0237	1.3024–40.3589
Neutropenia	1.4074	0.6911	0.2608–7.5954	6.0526	0.3752	0.1131–323.8045	1.4074	0.3304	0.1353–382.0527
GCSF	5.8696	0.3822	0.1108–310.8577	135	0.0455	1.1034–16517.8713	5.8696	0.6911	0.2608–7.5954
Admission	1.9333	0.6915	0.0745–50.184	48.3333	0.073	0.6961–3355.8421	1.9333	0.3822	0.1108–310.8577
Diarrhea	7.8889	0.1565	0.4530–137.3803	71	0.0141	2.3629–2133.3722	14.2	0.6915	0.0745–50.184
								0.0368	1.1772–171.2896

GCSF: Granulocyte-colony stimulating factor requirement; HFS: Hand and Foot Syndrome.

population-scale pharmacogenetic maps can be a cost-effective handy guide for clinicians to determine the type and class of treatment. Pharmacogenetic profiling of variations responsible for drug metabolism and availability is a huge step toward personalized medicine. Although patient level precision is not achieved, this method could still benefit in resource limiting conditions [56]. A comprehensive landscape of *DPYD* variants could be lifesaving and can enhance the quality of life for many patients. Patients with severe toxicity toward 5-FU often discontinue the drug and are also hospitalized in most cases. This could largely affect the patient's prognosis and quality of life [52]. Patients undergoing 5-FU treatment must undergo prior pharmacogenetic screening for *DPYD* mutations to find their susceptibility toward developing adverse effects for the drug.

For this study, we coalesced seven datasets from various publicly available resources to make a compendium of resource for SAGE and profiled for the allelic and genotypic frequency distribution of DPD variants that determines the bioavailability of 5-FU. We found that the allelic frequency of rs115232898, rs67376798 and rs55886062 are relatively low [35,37,40] and were not available in any of the dataset other than the 1000 Genomes. The presence of only the dominant alleles of rs75017182, rs1801158, rs12132152, rs17376848 and rs76387818 in the south Asian populations negate them from being a causative of 5-FU toxicity in the individuals from these populations. The heterozygous form of rs1801265, rs1801159, rs1801160, rs2297595 and rs12022243 are found to be more prevalent in almost all of the cohorts in south Asia. The presence of the allele rs1801265, rs1801159 and rs1801160 have been reported to have no marked DPD deficiency in Chinese cancer patients by He *et al.* in spite of their higher prevalence. CPIC guidelines have mentioned that the effects of these variants on DPD activity are contradictory [57]. Interestingly we found the homozygote recessive allele of rs1801265 to be predominantly present in the Andaman population, this could be because of a founder mutation that has now dissipated in the population.

We observed high occurrence of rs2297595 among south Asians, which is a contributor to toxicity in 5-FU administered patients. CC and CT genotype of the variant would have higher levels of 5-FU toxicity and hence should be avoided 5-FU administration completely. In our clinical study, we observed the prevalence of seven variants our of 15 DPYD pharmacogenetic variants tested. We found that three variants rs1801160, rs1760217 and rs12132152 to be highly prevalent in the study cohort, but did not show any significant correlation to the toxicity observed in the patients. Variant rs2297595 was found in 17 patients with CT genotype and two with CC genotype out of 110 patients screened, they also exhibited classical 5-FU toxicity symptoms like hand–foot syndrome and neuropathy. The odds of neutropenia, hand–foot syndrome and diarrhea were significantly higher with the heterozygote phenotype; and the homozygotic condition of this allele had all the classical side effects with significant p-value. Variant rs3918290 which has been reported as a causative of 5-FU toxicity is found in very low frequency in SAS population. We found two patients with a heterozygote phenotype to this allele and they exhibited grade III HFS and diarrhea that are the classic symptoms of 5-FU toxicity. It is also observed that Europeans and Americans are carriers of the risk allele in either form implying the importance of a genetic testing before the administration of 5-FU.

Comparison of allelic distribution of the 15 variants in south Asian population with the global populations revealed remarkable insights. Variants rs1801160 and rs12022243 are observed in higher frequency in south Asia compared with the rest of world [31]. Variants rs12132152 and rs76387818 have a similar allelic distribution among south Asians like in Africans. While variant rs17376848 is similar to Europeans, rs2297595 and rs1801265 showed affinity to both Americans and Europeans. These findings suggest that there is a lot of heterogeneity within the samples.

We uncovered 17,067 new polymorphisms in south Asian population by novel variant analysis. We identified 14 variants that were deleterious in nature by three independent functional analysis tools and hence could be potentially hampering the enzyme activity and thereby can cause 5-FU toxicity. These results will need to be confirmed by additional clinical studies.

Limitations of the study

Though the present study caters to the need for a genetic testing for cancer patients undergoing chemotherapy with 5-FU, the study could not cover all the variants that are related with the drug toxicity. Low frequency of genetic markers in a population impedes the real-world application of pharmacogenetics.

Conclusion

The present study provides the pharmacogenetic landscape of 5-FU in India and south Asia encompassing 15 variants of *DPYD* gene. This study reports a paradigm for understanding the interpopulation differences in 5-FU response and toxicity. Our study shows the variants of *DPYD* associated with 5-FU toxicity in different populations of Indian subcontinent and differences in the allele frequencies of Indians compared with the global population. Our analysis recommends the importance of genotyping the variants in cancer patients before chemotherapy of 5-FU.

Future perspective

DPYD variants show higher frequency and hence increased toxicity in cancer patients undergoing chemotherapy with 5-FU. Exome sequencing studies, whole genome sequencing studies and in-depth analysis of pharmacogenomics markers have the potential to identify novel rare variants with significant odds ratios in these populations.

Supplementary data

To view the supplementary data that accompany this paper please visit the journal website at: https://www.futuremedicine.com/doi/suppl/10.2217/pgs-2017-0101

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Summary points

Background
- Nearly 40% of cancer patients develop toxicity upon 5-FU administration.
- The adverse drug reaction toward 5-FU is accounted by partial or complete deficiency of dihydropyrimidine dehydrogenase enzyme.

Materials & methods
- Allele frequency distribution for 15 variants in *DPYD* gene was calculated among all the south Asian population from 3140 samples from seven datasets.
- Clinical validation was done for four variants on a limited cohort of 110 cancer patients.

Conclusion
- Our data suggest that variant rs2297595 is prevalent in south Asia with a higher allele frequency.
- Presence of this variant in heterozygous and homozygous form showed various toxicity-related symptoms associated with 5-FU adverse reaction.

manuscript or decision to publish. The authors have no other relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript apart from those disclosed.

No writing assistance was utilized in the production of this manuscript.

Ethical conduct of research

The authors state that they have obtained appropriate institutional review board approval or have followed the principles outlined in the Declaration of Helsinki for all human or animal experimental investigations. In addition, for investigations involving human subjects, informed consent has been obtained from the participants involved.

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